

The Role of Design Education Today

At a time when rapid changes in the landscape of technology are challenging traditional notions of the purpose and process of higher education, it is important to commit to the ideal of helping students learn and cultivate skills—to study, analyze, make, critique—that they can hold on to, finding meaning in, use to make sense of the world, and wield to bring about positive change. This is particularly necessary in the case of design education, where we need to urgently resist the ceding of both the craft work of design and the critical thinking that goes into it. Through all of my teaching experiences across undergraduate and graduate classrooms, engineering and design schools, and in-person and online instruction, I have found that students learn best when they are creating things—stories, prototypes, experiences—that enable them to express their hopes for the present and future, while viewing themselves as capable of action at both timescales. These ideas strongly align with the goals of the Keller Center, and I would be very excited to help contribute to its emerging teaching program in design. Before sharing some key vignettes from my teaching experience, I will outline two ideas for courses that I would be interested to teach. For each of the ideas below, there are many courses across the three minors at the Keller Center that already overlap very well, and I will mention some of them as we go along.

Ideas for Courses at the Keller Center

1. Conversations in Critical Theory for Design: Over the past year, I have been working my way through an opportunistic syllabus¹ broadly organized around “Critical Theory of Technology”, together with foundational readings in Science and Technology Studies, and more recent writings about critically responding to AI and economic overgrowth. I would be very interested in teaching a seminar course that discusses these readings in contrast to the dominant narratives of technological determinism, and places them in conversation with student attempts to express and justify their visions of a good life through design. I was able to pilot this idea earlier this year through guest lectures in a course on “Technical Communication and Design”. I was invited to speak to a class of engineering undergraduate students who were learning about “good” design as described in “The Design of Everyday Things” by Don Norman. I asked the class to read “Technologies as Forms of Life”² by Langdon Winner, and used it to lead a discussion³ on what a more critical “philosophy” of everyday things might mean for their work. The student response to this juxtaposition was quite encouraging, and I think a more full-fledged version in the form of a course—alongside existing ones like “Transformative Design” and “Technology and Society”—would potentially be a useful addition to the design minor.

2. How to be Generative without AI: The idea of this course is to take an explicit stance against Generative AI’s encroachment into design, and help students develop the skills to (1) build prototypes “by hand” across the physical and digital world, and (2) construct and navigate design spaces by generating a collection of meaningful and interrelated ideas that help form the basis of well-considered tools. As I will discuss in the next section, I have over time developed strategies to help students with little to no prior experience in more “technical” prototyping gain the confidence to see themselves as builders. In my own research, I enjoy the act of drawing from diverse disciplines to construct frameworks and spaces that guide design, and I have collected conceptual tools and processes that can help others do the same. Both “Creativity, Innovation, and Design” and “Design Fundamentals” seem to share many of the same goals and topics as this course idea, and it would be great to learn from the instructors to design a syllabus that can best complement these classes.

¹A list of some of these readings is here: rishivanukuru.com/critical-readings-26

²Technologies as Forms of Life by Langdon Winner. doi:10.1057/9781137349088_4

³An excerpt from the slides used for this discussion is here: rishivanukuru.com/phil-design-26

Some Vignettes of Teaching and Learning

My first formal teaching role was a teaching-assistantship (TA) position during the senior year of my undergraduate program in Civil Engineering. I conducted weekly tutorial sessions for the introductory course on “Engineering Mechanics”, and this formative experience helped me develop the skills of translating and communicating complex concepts in ways that are easy to grasp. Alongside my major, I also completed a minor in Design, which led me to switch pathways and enroll in the Interaction Design graduate program at my university. Most classes in this program operated in a “studio” model of learning, where we would spend a few weeks focused on a single module, learning both theory and practical skills across the domains of interaction, visual, and product design by working on self-driven collaborative projects with a strong focus on addressing key societal issues. In my second year in the program, I was first an “apprentice” TA, and then the lead TA for two graduate courses on Immersive Media Design. In these courses, I helped designers and artists with minimal software prototyping experience become comfortable with building projects spanning the range of mobile- and headset-based spatial computing. Having just gone through the learning-by-doing experience myself, I created a series of approachable development sessions to teach key concepts, and closely worked with project teams to help them bring their ideas to life. I continued to help with these courses as a guest lecturer for a few years post graduation, designing resources for hands-on sessions that work well in remote instruction settings.

I brought my learnings from all of these experiences together when I served as a Co-Instructor for the course “Introduction to Virtual Reality” during my PhD. This was a class consisting of 45 students from two programs—Computer Science and Creative Technology & Design—at both graduate and undergraduate levels. I was responsible for all lectures relating to the design and development of spatial computing applications. The course was offered in virtual attendance mode, and as a result, I had to speak to a mixed student audience with different backgrounds, while also ensuring an engaging and smooth remote learning experience. To achieve this, I designed multiple self-contained lectures that would take students from a blank window to a fully-functioning project in one session. I created technology-specific sandbox repositories for projects, with the necessary code and templates to quickly get started. These sandboxes functioned as carefully-designed “black boxes” that students could look inside, learn from, and extend as their understanding of the technology grew. This represents a process quite different to the current use of Generative AI tools to create projects, where the task of looking inside the black box and asking why things are a certain way often leads to more frustration and less real learning. To complement these online lectures, I led a team of Learning Assistants and held multiple in-person office hour sessions each week, where I mentored student teams and helped them hone the skills needed to fulfill their own creative vision. In the qualitative course feedback, students particularly highlighted the benefits of the structure, delivery, and project-oriented focus of the lectures.

In the years since, I have continued to help students and teams work their way through projects that span a range of technical domains, and have found that the best way to counter the Generative AI-driven offloading of craft is to make the process of building “by hand” more approachable and encouraging. I believe that these experiences, together with my own ongoing critical learning journey, place me in good stead to help shape and actively participate in the teaching program at the Keller Center.

On a more personal note, my thoughts about teaching have been significantly influenced by my mother, who is a high school mathematics teacher with nearly thirty years of experience. For a ten-year period during my school and undergraduate education, she conducted after-school classes for students in our home. She is particularly good at helping students build confidence in what can be quite a frightening subject, and I have seen first-hand the impact of kindness, patience, and meeting students where they are. I hope to continue learning from her as I try to emulate her enthusiasm, understanding, and resilience in my own teaching.